Precision Routing for Efficient Robot Policy Deployment on Edge Devices

CS 2420 Project Proposal

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Problem Statement

How can robot policies be deployed efficiently on compute-limited edge devices while achieving both a high success rate and low latency?

Motivation

Deploying reinforcement learning (RL) policies on robots is often bottlenecked by limited on-device compute. Quantization reduces this cost but can degrade performance. However, not all states demand the same precision: coarse motions tolerate errors that fine manipulation does not. We therefore propose to learn a routing framework that dynamically selects which precision model to use at each step.

Proposed Approach

We will post-train multiple quantized versions of a robot policy at different bit-widths. We will then train a router in one of two ways:

(1) **Supervised learning** on a labeled dataset generated by running the high-precision and quantized models on the same inputs.

(2) **Reinforcement learning**, treating the router as a policy.

Evaluation

We will evaluate the resulting system on:

- task success rate,
- inference latency,
- energy usage, and
- roofline analysis.